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**Specific contributions of cortical cortex on the integration of
audio-visual walking inputs in patients with stroke compared to
healthy subjects: an EEG frequency-tagging approach**

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Abstract: The brain dynamics underlying sensorimotor synchronization constitute a complex combination of different neural processes. Here we wanted to investigate the contribution of cortical cortex on rhythmic sensorimotor synchronization when coupling with auditory and visual inputs related to walking movement. The study has a mixed-models study design with auditory (footsteps sound) and visual (point-light figure) inputs set at the frequency rate of 2 Hz. They were used in three different blocks: audio, audiovisual and visual. The study was conducted by means of electroencephalography (EEG) in a frequency-tagging approach on two distinct groups of individuals: healthy control subjects ($N = 21$) and patients with a stroke ($N = 21$). The results of both population groups showed neural responses of the sensorimotor area in all the stimuli presented in a rhythmic condition, though not in the random one. Such outcome is sustained by behavioral questions on the positive feeling towards the distinct stimuli, which appeared with significantly higher values for the inputs repeated in the rhythmic condition. Moreover, sensorimotor synchronization appeared in auditory and audiovisual inputs, but not in the visual one. In the stroke population the neural response patterns resulted weaker than in the control group, and left-lateralized. These results demonstrate that the sensorimotor synchronization can be elicited only by sensory stimuli repeated in a rhythmic sequence and that auditory inputs appear better to induce neural entrainment. Furthermore, it could be concluded that stroke patients preserved the ability of sensory integration and neural entrainment, even if it results reduced compared to healthy individuals.

Keywords: EEG, Frequency-tagging, Stroke, Sensorimotor integration

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1. Introduction

An average of 5.5 million people worldwide die every year because of a stroke, ranking it as the second leading cause of death and one of the main cause of cardiovascular diseases (Donkor, 2018). Research in the United States has stated that more than 795,000 people have a stroke every year, that is almost one person every forty seconds (Tsao et al., 2023). In the past three decades, an increase in the incidence of stroke was globally observed, especially in young (18-44 years old) and mid-life (45-64 years old) populations, yet not in older adults (over 65); on the other hand, the incidence of death and disability has decreased as a result of the advance of research and medicine (Yahya et al., 2020). Even though a decrease in disability has been observed, stroke remains the leading cause of serious long-term disability: more than 50% of people who survived a stroke become chronically disabled, both in neurological and physical aspects. One of the most common consequences is long-term motor impairment due to lesions in cortical or subcortical motor areas (Karthikeyan et al., 2019).

Stroke is divided in two main subtypes, drawn by the cause of it. The most common one results to be the ischemic stroke, it's produced by a blockage of blood vessels and comprehends TIAs (transient ischemic attacks) or mini strokes, which only last for a few minutes up to 24 hours. The second type is the hemorrhagic stroke which is caused by bleeding in or around the brain, and it appears to cover just the 15% of the cases (Abdu, Tadese, & Seyoum, 2021). On the other hand, if we look at the brain regions lesioned by stroke, it's possible to differentiate it between cortical and subcortical injuries, which can alter different cognitive (e.g. language, space orientation) and physical abilities (e.g. motor impairment, weakness), depending on the specific cerebral area and lobe affected.

One of the major consequences of experiencing a stroke is impaired gait: a total of 80% of patients have an impaired walking ability, with 50% of patients who are completely unable to walk, 12% who can walk with assistance, and 37% can walk independently (Balaban & Tok, 2014). In those cases, gait can result as hemiplegic, slower in speed and cadence and as well as shorter stride length (Gomez-Andres et al., 2020). The balance of the locomotor control is shift due to the lesion in the central nervous system (CNS) affecting automaticity, proprioception, balance, anxiety state and multisensory feedback, which will impair walking abilities (Clark, 2015).

Due to the wide spreading of gait impairment in patients who suffered from a stroke, the recovery of walking ability is of the main aims in stroke rehabilitation. Recently rehabilitation techniques have been moving towards top-down approaches, which base the rehabilitation on the state of the brain after stroke. These techniques use a motor learning approach mainly through the integration of sensory feedback, helping the proprioception of the body thanks to sensory-motor adaption (Lois et al., 2011). The human brain presents the function of multisensory integration: it integrates the information of different sensory modalities into a coherent representation (Stein & Stanford, 2008). Multisensory integration can improve motor responses, through the ability of the brain to extract sensory information regarding body position and anticipate future positions. Additionally, it has been observed that many multisensory processes appear to be largely preserved in stroke patients, keeping integrate the ability to synthesize multisensory information representing motor activity (Bolognini et al., 2012).

From one side, research has investigated the usage of different sensory inputs for post-stroke gait rehabilitation: for what concerns visual stimuli, it has been shown that Mirror Neuron System (MNS) play a crucial role in action observation and motor learning, through the stimulation of an internal model, able to reinforce sensorimotor representation (Rizzolatti & Craighero, 2004). Furthermore, visual feedbacks used in motor imagery-based training can also improve motor recovery in post-stroke patients, especially the ones affected by severe hemiparesis (Mihara et al., 2013). On the other hand, auditory feedback has been a fundamental sensory feedback for gait rehabilitation, being used as a real-time cue to correct movement and redirect walking ability. Sounds can activate excitability through auditory-motor circuits at the reticulo-spinal level; if they are repeated rhythmically, they can entrain the timing of muscle activation, enhancing movement during rhythmic actions (Thaut et al., 1999). Lastly, the employment of audiovisual stimuli has been explored: MNS appears to respond to the visual representation of action, as well as the auditory one (Rizzolatti & Craighero, 2004); the integration of both sensory modalities can result more beneficial considering the positive effects of multisensory integration, both in healthy and clinical populations (Bolognini, Russo, & Vallar, 2015). Both auditory and visual stimuli have been employed in different neurophysiological rehabilitation techniques, such as in Proprioceptive Neuromuscular Facilitation (PNF) (Moros et al., 2000) or Motor Imagery (MI) (Mason et al., 2007).

Some evidence has found auditory feedback as fundamental for walking rehabilitation: it reacts faster than the visual system and can continuously be delivered without constraining the movements. From such discoveries, movement sonification has been engaged for these goals; here the physical properties of the human movement are being translated to audio parameter (e.g. the speed or position of the body are translated into volume or timbre of audio sounds) (Effenberg et al., 2016), enhancing motor learning based on motor perception and its representation (Bevilacqua et al., 2016). Still, little research on movement sonification for sensorimotor learning and movement recovery has been done, even more the study of the neurological processes underlying such phenomena in clinical population.

Sensory feedback employed for movement rehabilitation induce biological motion perception. Studying the neurological processes that underline motion perception, results as a complex goal, since multiple social cues perception and process are involved. In order to isolate only the interested brain activity related to motor perception a new paradigm studying neural entrainment through electroencephalogram with a frequency-tagging approach, with a neutral visual stimulus of a point-light figure walking, has been proposed by Cracco et al., 2022.

Neural entrainment or sensorimotor synchronization is defined as unidirectional synchronization of neural oscillations to an external rhythmic stimulus (Lakatos, Gross, & Thut, 2019; Haegens & Golumbic, 2018); an action is rhythmically coordinated with a predictable external stimulus (Pressing, 1999). We use rhythmic synchronization between the auditory and motor systems in many of our everyday behaviors, one of the main being speech. Our brain leans towards a natural synchronization of its movement with endogenous rhythmic signals: auditory rhythms rapidly entrain motor responses into stable steady synchronization states (Thaut, 2003). Such alignment of motor and sensory rhythm at the neural level can be detected in music, dance, sports, rhythmic tasks and verbal communication (Rosso et al., 2023). Sensorimotor entrainment can be driven by a status of phase- and frequency-locking of neural oscillations,

which will reflect the changes in the external sensory through neuronal excitability (Lakatos et al., 2005).

A way that has been found successful to study neural entrainment is the frequency-tagging approach, where a brain response is elicited through a repeated cycle movement; if the aim is movement perception related to walking the brain response can be elicited at the end of every footstep (Cracco et al., 2022). The cognitive and sensory processes present in neural entrainment (brain synchronization with the sensory input presented) are represented by neural oscillations that can be observed through electroencephalography (EEG): EEG signals result in time-locked to re-occurring sensory stimuli (Thut, Miniussi, & Gross, 2012). State-of-the-art approaches tag frequencies of a rhythmic stimulation in the power spectrum of EEG in order to quantify neural entrainment (Nozaradan et al., 2011): the "frequency tagging" methodology converts brain signals into their frequency components.

The usage of frequency tagging with electroencephalography results in the most appropriate technique to study the role of different brain regions integrating multisensory stimuli through neural entrainment, being able to isolate the brain processes related to movement perception, distinguishing them from other cognitive processes that may be related to social cues using neutral sensory inputs, such as a point-light figure with no gender, weight nor emotional trait (Cracco et al., 2022). Brain synchronization remains a complex phenomenon, previous research showed that listening to repeated rhythmic beats activates cortical oscillation evoked from auditory stimuli and anticipations of it (Snyder & Large, 2005). The different brain regions that are being enhanced during auditory-motor entrainment include: the cerebellum (Grahn, Henry, & McAuley, 2011), inferior colliculus (Tierney & Kraus, 2013), basal ganglia (Thaut et al., 2009) and cortical hubs, located at the top cortical networks, that enable sensorimotor integration. In conclusion, research proved that the auditory and visual feedback can elicit motor responses by neural entrainment in the motor cortex, generating synchronized brain oscillations to the rhythm of the stimuli; furthermore different studies demonstrated that such feedbacks play a key role in stroke gait rehabilitation (Chen et al., 2018; Bolognini, Russo, & Edwards, 2016), facilitating motor automaticity, redirection and action.

Even if more research in gait rehabilitation has been done in patients with stroke, few investigations have examined the neural activation related to the integration of multisensory stimuli (audio and visual) related to walking movement. Hence, the aim of the present study is to investigate the brain dynamics supporting the sensorimotor synchronization when coupling auditory (footstep sounds) and visual (walking point-light-figure) inputs (see Figure 1) related to walking movement in a group of patients with stroke compared to a group of healthy subjects. The study will be conducted using an electroencephalogram frequency tagging approach, aiming to elicit neural synchronization between brain oscillatory activity and the presented sensory inputs (visual: walking point-light-figure; and auditory: footstep sounds) related to walking ability presented at 2 Hz. The frequency rate presentation of the stimuli was selected based on previous study conducted in our laboratory by Matamala-Gomez et al. in which the authors observed that at 2 Hz when presenting auditory and visual inputs at rhythmic sequences there was a higher synchronization with the sensorimotor cortex in healthy subjects (Matamala-Gomez et al. in preparation). In detail, in the previous study the authors compared different frequency rates on the presentation of the sensory inputs (1, 2, and 3.6 Hz) in rhythmic and random sequences, and observed a higher synchroniza-

tion with the temporal and sensorimotor cortex when presenting auditory (footstep sounds) and visual (walking point-light-figure) at 2 Hz in rhythmic sequences. Moreover, the authors also observed a higher synchronization with the Occipital cortex when presenting the visual input (walking point-light-figure) at 2 Hz in rhythmic sequences. Based on the previous results, in the present study we expect to find a higher brain synchronization when presenting auditory inputs (footstep sounds) at 2 Hz in rhythmic sequences, with the temporal and sensorimotor cortex. Moreover, we also expect to find a higher synchronization when presenting the visual inputs (walking-point-light figure) at 2 Hz in rhythmic sequences with the occipital cortex and a weaker elicitation of the sensorimotor cortex. For all the sensory inputs we predict to see a significant higher neural synchronization in the stimuli a rhythmic sequence compared to the random sequences. Finally, regarding the patients with stroke, we expect to observe some differences in brain synchronization compared to the healthy control group. Since cortical and subcortical areas were depicted as facilitators for motor alignment thanks to the ability of beat-based time-keeping (Cannon & Patel, 2021), we expect differences of brain activation in stroke patients compared to healthy subjects, but still, visible neural entrainment elicited by repeated rhythmic multisensory stimuli.

2. Methods

2.1 Participants

In the present study, we had a total population of 42 subjects, equally divided into two groups: 21 individuals were part of the healthy control group, whilst the remaining 21 were patients who suffered from a cortical or subcortical stroke. The recruitment for the control group was made through several methods: in person, talking to alumni of the Universitat de l'Experiència of Barcelona, through personal contacts, and using the platform Sonasystem. On the other hand, to recruit participants who experienced a stroke we used a database previously employed in another study made in the Cognition and Brain Plasticity Unit for movement rehabilitation with music therapy. Furthermore, we were also able to present our study at Fundació Ictus, where we could select more participants. The exclusion criteria in both groups were: no visual or auditory impairments and no pregnancy.

Participants of the two groups were matched by age and gender, where we can find a mean of 55,4 years old in the control group and of 56,4 for the stroke patients, with a non-significant difference (t test: -0.299, p value: .766). Likewise, the difference in gender did not result significantly different (t test: 1.56, p value: .127), even though higher participation of men compared to women can be observed in the stroke group and a preponderance of women compared to men can be observed in the control group. To both groups a questionnaire measuring the level of physical activity was administered: an average of 7026,78 kilojoules a day was found for the control group, whilst for the stroke group it resulted of 5669,19 kJ/d, which results lower than the control group, but still without a significant difference (t test: 1.06, p value: .146). Regarding the stroke group, the clinical history of the patients was collected, founding the majority of patients with the injured left hemisphere, which affected their contralateral right extremity ($N=13$ suffered the injury in the left hemisphere; $N=7$ in the right hemisphere; $N=1$ in both hemispheres; a database can be found in Table A.2). Participants signed an informed consent before the experiment, and they were reimbursed for their time.

2.2 Materials

The experiment was conducted with electroencephalography, recorded using a standard set-up with 64 Ag-AgCl electrodes placed on the scalp according to the International 10/10 system (EasyCap) to capture the neural activity in the whole brain. The study presented auditory (footsteps sounds) and visual (point-light-figure) inputs (Figure 1) related to the walking ability, all set at the frequency of 2 Hz, using Matlab for the visual stimuli and Audacity software for the auditory ones; resulting the best to activate neural entrainment (Matamala-Gomez et al. in preparation).

For what concerns the visual stimuli, a point-light figure walking, representing the human body through white dots set in the primary joints of the body (Figure: 1), was used: it was created with BioMotionLab, using a neutral subject, without any specific gender nor emotional display, with an average weight. After being created with the mentioned software, its frequency was adjusted to the desired one.

The choice of a neutral point-light figure as the visual stimulus was made to focus only on brain activity elicited by movement processing frequency. Earlier studies have in fact shown that the presence of different important biological and social features can be inferred from figures and may enhance other processes not primarily related to movement perception (Cracco et al., 2022). For the audio stimuli, the sound of footsteps at 2 Hz with a neutral connotation was engaged. The participants listened to the sound through headphones and they were asked to pay attention to the blue fixation cross displayed in the middle of the black screen. Finally, a third stimulus consisting of the union of the point-light figure walking and the sounds of the footsteps, was employed and denominated as audiovisual. Moreover, after

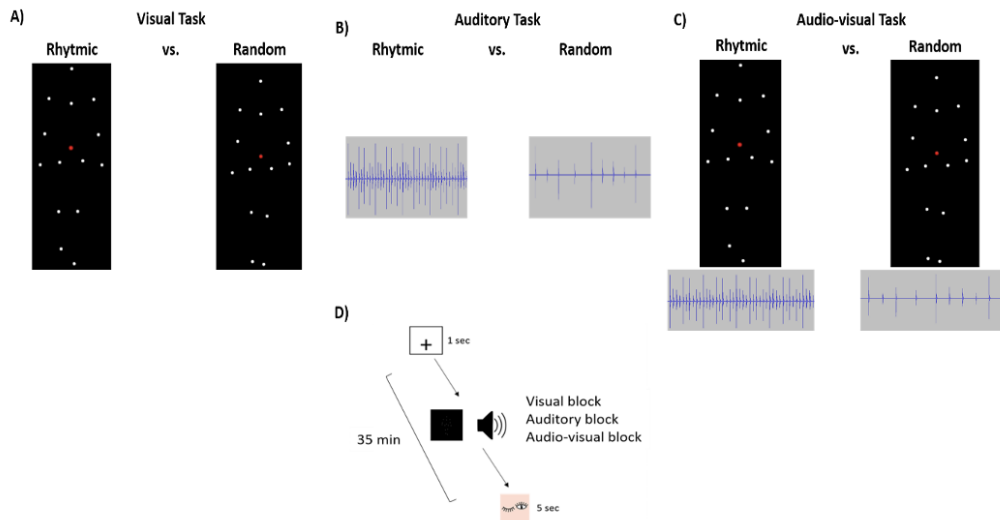


Figure 1: Auditory, visual, audiovisual inputs, and the experimental timeline

the observation of each sensory input, six subjective report questions were presented: they referred to how the participant felt while seeing the point-light figure walking or listening to the footsteps sound. The questions had to be answered on a Likert scale with values between 1 (strongly disagree) to 5 (strongly agree) (see the Experimental Design section for further details and (Table 1)). The experimental task was developed using with E-prime software, which allowed us to build an experiment with different stimuli presentations (e.g. adding images, questions and text).

To measure the level of physical activity of each participant, which would be later correlated to the EEG results, we used the *Active-Q* questionnaire (it can be found in the Appendix A.(A.1)), which was created precisely to assess the total physical activity and inactivity in adults, through a series of multiple choice questions (referred to a one-year time) on daily activity, mean of transportation used, leisure and sports activities; and the time a day spent for each of them (Bonn et al., 2012). The questionnaire was translated in Spanish from the original English version and transformed from a PDF file into an interactive digital version using the platform Qualtrics.

2.3 Experimental design

The current study is a mix models study design presenting both auditory (footstep sound) and visual (walking point-light figure) inputs at the frequency rate of 2 Hz in a rhythmic or random sequence, in three different blocks (auditory, visual and audiovisual). The blocks were presented in a randomized order to all the participants. At the beginning of the experimental session, we included a text with the instructions and right afterwards a white fixation cross preceding each sensory input presentation. In each experimental block the sensory inputs were presented four times in rhythmic sequences and four times in random sequences, each sensory input presentation lasted 60 seconds. The sensory inputs were presented in rhythmic and random sequences in a random order within each experimental block.

During the presentation of each sensory input an attentional task was set. For the auditory inputs, the participants had to detect changes of the pitch (to a more acute or grave footstep sound) in the footstep sound stimuli. For the visual and audiovisual inputs, the participants had to focus their attention on the central dot of the point-light-figure which changed from red to white. The participants had to report to the experimenter the number of changes in both pitch footstep sound and the color of the central dot after each sensory input presentation. Moreover, after each sensory input presentation the participants had to answer six subjective report questions in a 5-points Likert scale (see Table 1). Finally, after answering the subjective report questions, an eye-blink figure was presented during fifteen seconds indicating that they had time to blink if needed (see Figure 1 for the experimental block design). A break of roughly five minutes was taken between all blocks to fix the conductivity of the electrodes with a gel ¹ and to let the participants rest. Altogether, the experiment and the set-up of the EEG cap on the participants had a duration of almost two hours.

For what concerns the Active-Q questionnaire, it was administrated via e-mail a few days before the experiment, together with some additional information about the study.

2.4 Procedure

Before the experiment began the participants had to sign the informative consent, and they would ask questions if needed. Afterwards a careful preparation was made, where we placed the EEG cap on their head², the external on the side of the right eye and under it, as well as on the mastoids. Then a gel was inserted in each electrode (both external and placed on the cap) in order to diminish the impedance and allow a good signal connectivity. Additionally, if the participants had any trouble completing the Active-Q questionnaire at home, they were helped to do so before the start of the experimental session.

The participants sat on a chair in front of the computer screen where the stimuli were displayed and wore headphones to listen to the auditory stimuli. They were asked to avoid moving and blinking when the sensory stimuli were presented to avoid noise signal in the EEG recording. After reading the instruction on the screen, the participants had to pay attention to the stimulus that was being presented

¹The gel employed in the EEG procedure is a mix of salts and water that allows the conductivity between the scalp and the electrodes.

²The EEG cap was previously prepared with all the 64 electrodes in their relative site.

and count the number of changes in the color of the point-light figure's central dot or in the tone of the footstep sounds. At the end of each stimulus, they were asked to tell the experimenter how many changes they were able to observe or hear through a speaker³.

Afterwards, the subject questions were presented and had to be answered by the participants using numbers from 1 to 5 on the keyboard placed in front of them. When a question was answered the following one was automatically shown. When all the six questions were completed the eye-blinking image was presented, this was inserted for the participants to yawn, blink or stretch their face muscles if needed; furthermore they were expressively told that they could do so also during the answering of the questions, since the EEG recording during those time was of no interest for our analysis.

After all the eight stimuli with their relative questions were shown, a block was completed and a small break took place, where the participants were offered something to drink or eat and the electrodes' impedance was checked. When all the blocks were completed, the participants could wash their head from the gel in a sink.

Subjective report assessment scale

For the subjective report questions presented at the end of each experimental block we used six question related to the participant's perception of the sensory inputs, which can be found in Figure 1. We inserted these questions to understand the feelings produced by the sensory inputs; in particular we aimed to see if the perception of involvement with the sensory input or the positive feelings recognized during the presentation of such stimuli, could influence the level of neural entrainment related to that particular sensory input. Further, we wanted to understand if the difference of sensorimotor synchronization previously observed between the presentation of the stimuli presented in a rhythmic or random sequence (Matamala-Gomez et al. in preparation) could also be confirmed by a subjective report on their perception. For these purposes, we decided to insert two questions on the involvement of our body with the sensory stimuli presented (Q1 *«I felt involved with the marching movement of the white dot figure during the video observation (it was as if I was also moving)»* and Q4, measuring agency *«During the observation of the video, I felt like I was inside the body of the white dots (as if my own body was performing the movement)»*); two questions on the positive feelings evoked by the stimuli (Q5 *«During the observation of the video, I had a pleasant sensation»* and Q6 *«During the observation of the video, I felt relaxed»*); one on the fluidity of the stimuli (Q3 *«The movement of the white dot figure seemed fluid to me.»*); and one on the perception of the rhythm, which was considered the most significative one (Q2 *«The rhythm of the steps of the white dot figure seemed to reproduce the rhythm of my real march»*). All the questions were adapted for each different sensory input, and can be found in Figure 1. Their score had to be given on Likert Scale from 1 (strongly disagree) to 5 (strongly agree). The results obtained from the questions were collected into a *.txt* file by E-prime and inserted in an Excel table through a Python script, so that later they could be easily used to make statistical analysis.

³The speaker was used to communicate with the participants, which were in a different room from the experimenters. Furthermore, they could be seen through a webcam, and make signs if anything was needed.

Subjective Report Questions		
Visual	q1	I felt involved with the marching movement of the white dot figure during the video observation (it was as if I was also moving).
	q2	The rhythm of the steps of the white dot figure seemed to reproduce the rhythm of my real march.
	q3	The movement of the white dot figure seemed fluid to me.
	q4	During the observation of the video, I felt like I was inside the body of the point-light figure (as if my own body was performing the movement).
	q5	During the observation of the video, I had a pleasant sensation.
	q6	During the observation of the video, I felt relaxed.
Audio	q1	While listening to the sound of the steps, it seemed as if I was involved as if the sound reproduced my real marching movement.
	q2	It seemed to me as if the rhythm of the sound of the steps I heard through the headphones reproduced the sound of my real march.
	q3	The sound of the steps seemed fluid to me.
	q4	While listening to the sound of the steps through the headphones, it seemed as if my own body was producing the sound of the steps.
	q5	While listening to the sound of the steps, I had a pleasant sensation.
	q6	While listening to the sound of the steps, I felt relaxed.
Audiovisual	q1	I felt involved with the marching movement of the white dot figure during the video observation along with the sound of the steps (it was as if I was also moving).
	q2	The rhythm of the steps of the point-light figure along with the sound of the steps seemed to reproduce the rhythm of my real march.
	q3	The movement of the white dot figure along with the sound of the steps seemed fluid to me.
	q4	During the observation of the video and when listening to the sound of the steps, I felt like I was inside the body of the white dots (as if my own body was performing the movement).
	q5	During the observation of the video and listening to the sound of the steps, I had a pleasant sensation.
	q6	During the observation of the video and listening to the sound of the steps, I felt relaxed.

Table 1: Table: Subjective report questions translated in English from Spanish

Regarding the data for the Active-Q questionnaire (the questionnaire can be found in the Appendix A.1), we conducted the score calculation following the instruction reported on the article where it is explained, and its efficiency is tested (Bonn et al., 2012). Initially, we exported in an Excel file the results collected by Qualtrics with their numeric values (e.g. if the participant had selected the first option, the result would have been 1), this was imported in a Python file where the numeric values related to each activity were transformed into the MET values reported in Table A.1, which were encountered in the previously cited article. On the other hand, the frequency of each activity was measured multiplying the hours/minutes a day for the number of days per week; if a range of time was presented, we previously calculated the average of it, and then convert it into hours per day (e.g. the range 15-29 minutes/day was converted to 0.36 h/d). After creating an Excel table with the new scores accorded to MET values and the duration of hours/week, we proceeded to calculate the final Active-Q score, which returns a number

of kilojoules per day, using the following formula:

$$EE_{\text{activity}} (\text{kJ/day}) = MET_{\text{activity}} \cdot \text{Weight (kg)} \cdot \text{Duration}_{\text{activity}} (\text{h/d}) \cdot 4.184$$

The MET value and duration of the activity were multiplied by the weight of each participant (which was asked individually to the participant before the experiment), and the number 4.184, which is a conversion factor used to transform values from kcal to kJ. The obtained results represented the daily physical activity of each participant.

2.5 Analysis

2.5.1 EEG data analysis

The EEG data was registered through the software provided by BrainVision at the sampling rate of 1000 Hz; for all the subjects we made three different registrations, one per each experimental block (auditory, visual, and audiovisual). Before conducting the frequency-tagging analysis, the data was preprocessed using the EEGLAB toolbox ran with Matlab: all the chunks of register that weren't necessary (i.e. the initial part with the instructions, the one related to the eye-blink figure and the ones corresponding to the subjective report questions) were trimmed. Later we re-referenced the data to channels 31 and 32, which correspond to the electrodes TP9 and TP10, positioned externally on the mastoids. Finally, we ran the Independent Component Analysis (ICA) to remove blinking and muscles artifacts and saved the preprocessed data into a new file.

After the data for every block and for all the participants was preprocessed, we started the analysis with the help of some scripts made for the previous study and adjusted according to our necessities. Firstly, we extracted the epochs from the processed data: epochs of 20 seconds were obtained from the 60 seconds of each stimulus; the method of surface Laplacian and Current-Source Density (CSD) transform was applied to the data, dividing it between the one related to the rhythmic stimuli and to the random ones. From the files just generated, we computed spectral power activity with FieldTrip toolbox of Matlab, using *mtmfft* method, which allowed performing later analysis in time and frequency domains to study oscillatory brain activity and its modulation over time. The Fast Fourier transform (FFT) was applied to each sensory input (visual, audio and audiovisual) in both rhythmic and random sequences; therefore six variables were obtained.

Using these variables we computed the averaged Signal-to-Noise Ratio (SNR), calculating the averaged signal-to-noise ratio (SNR) and the mean power spectrum and its variance through the different frequencies. We also plotted the activity power spectrum to see how the neural synchronization to rhythm would change through frequencies, and to observe whether the desired peak amplitude activity at 2 Hz was present. For the activity power spectrum analysis we decided to take into considerations the different regions of interest (ROIs) of our experiment, concerning the Occipital, Temporal and Sensorimotor areas, respectively enhanced by visual and audio sensory inputs as well as motor synchronization. For the analyses, the three different ROIs included the following electrodes: Temporal (right: FT9, FT7, T7,

TP7; left: FT10, FT8, T8, TP8), Sensorimotor (right: F5, F3, F1, FC5, FC3, FC1, C5, C3, C1; left: F2, F4, F6, FC2, FC4, FC6, C2, C4, C6), and Occipital (PO3, POz, PO4, O1, Oz, O2).

Furthermore, in order to neutralize the effects that the side of the injury in stroke patients might have on our results, we decided to invert the location of the electrodes for the patients who suffered the lesion in right hemisphere, which were the minority. Doing so the topographies that were generated should result not affected by the different injured side of the brain and the left hemisphere becomes the Affected Hemisphere in the stroke group and the Right hemisphere would be the Unaffected hemisphere. The interpretation of the corresponding figures in the results' section should be in consideration this aspect.

Finally, we observed the distribution of the power spectrum activity for each sensory input, in both rhythmic and random sequences for each experimental block when presenting the sensory input at 2 Hz, in healthy individuals and patients with stroke. Moreover, we also could observe the power spectrum activity difference between rhythmic and random sequences (Figures 7 and 8).

2.5.2 Subjective report questions analysis

For the subjective report questions, we analyzed the mean difference between rhythmic and random sequences in each experimental block (auditory, visual, and audiovisual) for each subjective report question of the questionnaire. Further, we also calculated the mean difference between healthy subjects and patients with stroke in both rhythmic and random sequences in each experimental block for each subjective report question (Figure 10a). In specific, we used the Wilcoxon Signed-Rank test for non-normally distributed data or the paired T-test for normal distributed data. We tested the distribution of the data with the Shapiro–Wilk test.

On the other side, with the scores of the physical activity, obtained with the Active-Q questionnaire, we aimed to explore the relationship with the subjective report questions and the brain power spectrum. To that aim we used Spearman's correlation test for non-parametrical data. For the correlations, a numerical value for each participant and condition representing the mean power spectrum activity of each specific ROI (temporal, sensorimotor, and occipital) during each sensory input exposure (auditory, visual, and audiovisual) only at rhythmic sequences, was extracted, in both healthy and patients with stroke. In order to see whether there was a relationship between the power spectrum activity and behavioral responses, we selected the question Q2 (see Table 1), that was the one more related to the rhythmic perception of the sensory input presentation.

3. Results

3.1 EEG results

In this thesis, the EEG results will be presented in a descriptive manner, in which we describe the outcomes presented in the obtained oscillatory waveforms (Figures 1, 4) and topographies (Figure 7, 8). Further, statistical analyses will be conducted in the future. As expected from our previous study conducted by Matamala-Gomez et al. (in preparation), in our research we also observed a greater synchronization effect in rhythmic sequences compared to random sequences in all three ROIs (occipital, temporal, and sensorimotor) and in each experimental block (auditory, visual, and audiovisual). Moreover, as in the previous study, we also observed a synchronization effect in the sensorimotor cortex when presenting auditory and audiovisual sensory inputs in rhythmic sequences at 2 Hz, in both healthy population and patients with stroke (Figures 4, 5, 6).

However, visually, the topographies tend to show that mean power spectrum activity is lower in patients with stroke compared to healthy subjects. This tendency should be confirmed using proper statistical analysis. From the topographical representation of these effects (see Figures 7 and 8), we can see clear differences in the sensorimotor effects in the stroke patients compared to the control group. For example, in the Affected hemisphere in stroke patients (left-hemisphere view), the activation observed in the sensorimotor cortex in the patients is more lateral and posterior, showing less magnitude. Also in the audiovisual condition, the activation in the healthy hemisphere is less and being more anterior compared to the control group. For the visual sensory input, in our study we also observed a higher synchronization of the visual input within the occipital cortex in rhythmic sequences, but not showing entrainment at the sensorimotor cortex. This result was also observed in the previous investigation from Matamala-Gomez et al. Overall the present new results tend to show clear brain reorganization effects in the sensorimotor cortex, showing less entrainment and variability in these regions, probably due to the presence of a lesion in the motor system.

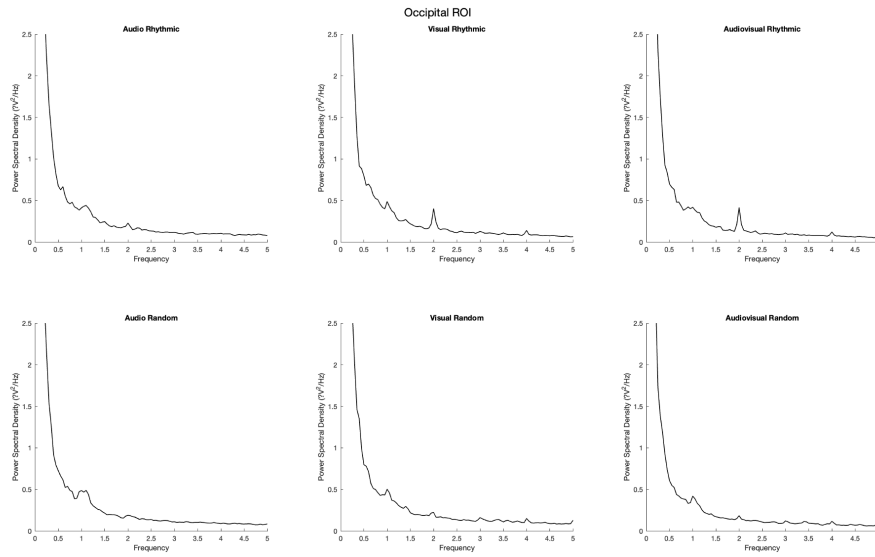


Figure 1: Power spectrum of the Occipital ROI: control group

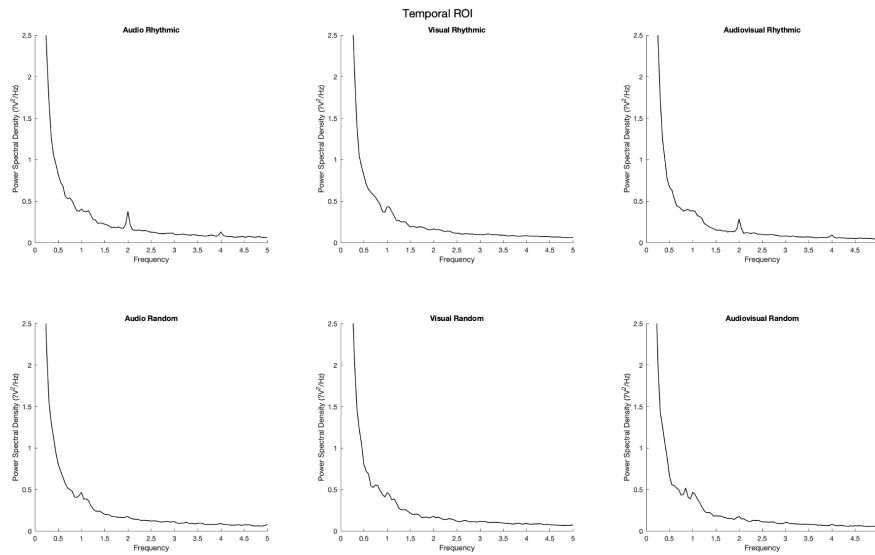


Figure 2: Power spectrum of the Temporal ROI: control group

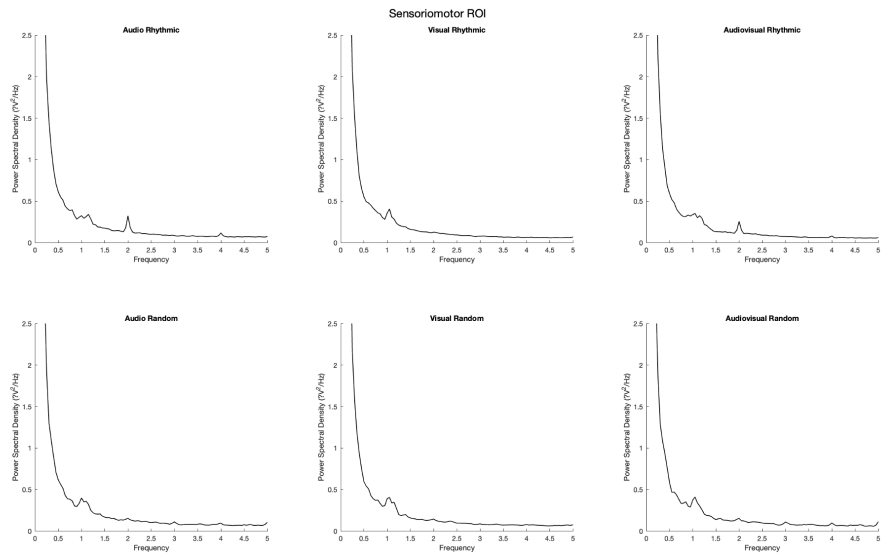


Figure 3: Power spectrum of the Sensorimotor ROI: control group

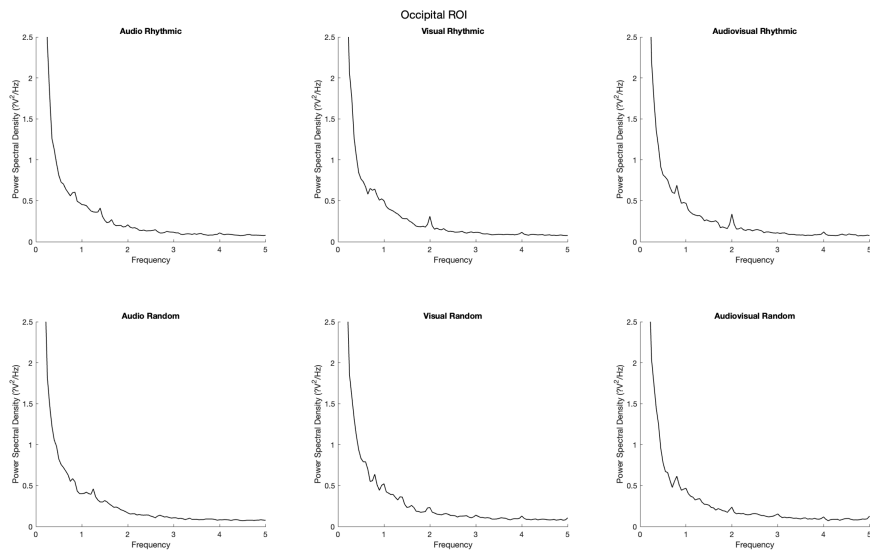


Figure 4: Power spectrum of the Occipital ROI: stroke group

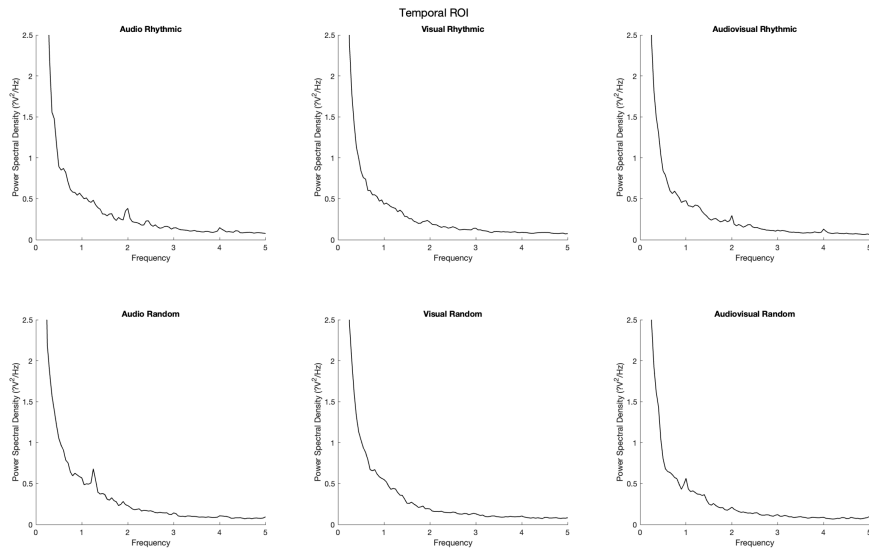


Figure 5: Power spectrum of the Temporal ROI: stroke group

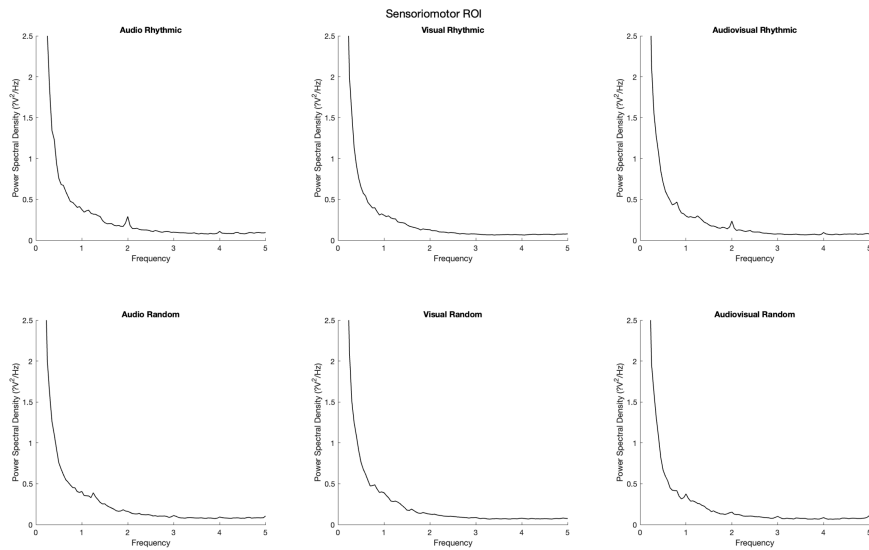


Figure 6: Power spectrum of the Sensorimotor ROI: stroke group

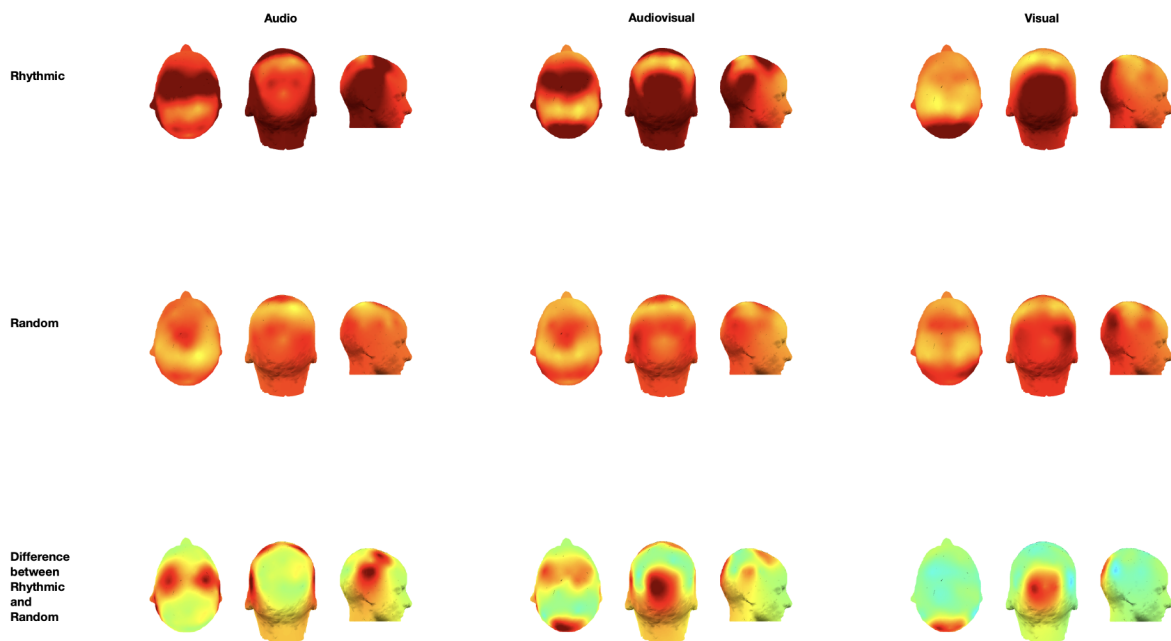


Figure 7: 3D topographies related to the activity in the control group

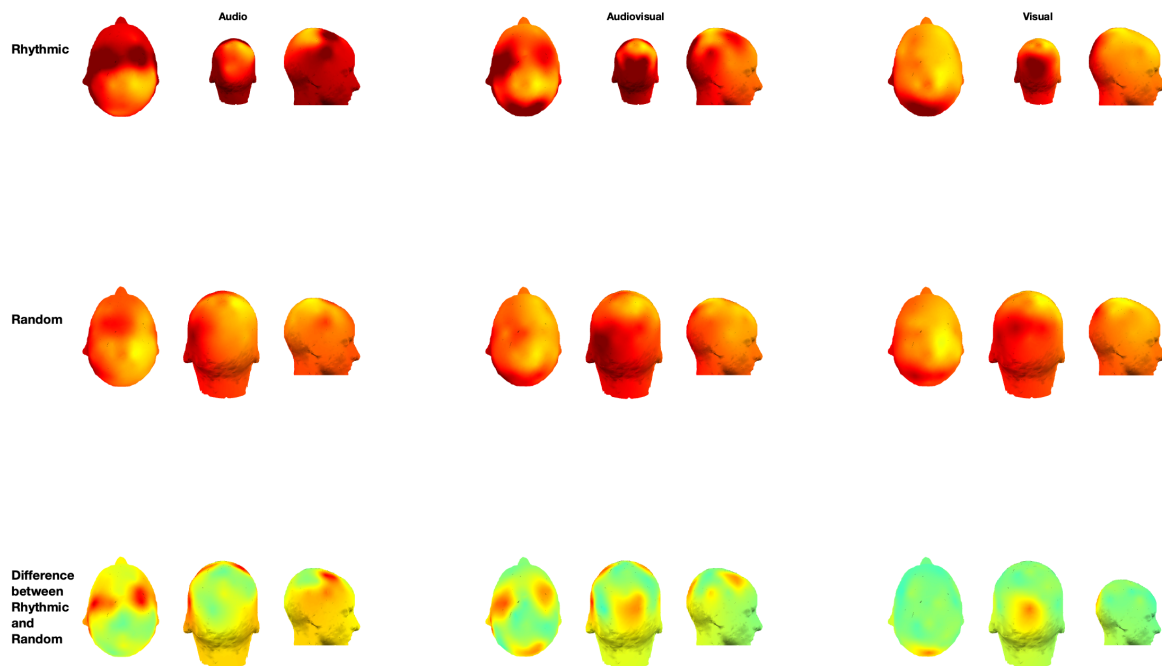


Figure 8: 3D topographies related to the activity in stroke population

3.2 Subjective report questions results

In the subjective report questions results, we calculated the mean difference between the values obtained from the subjective questions provided after the presentation of each sensory inputs in each experimental block (auditory, visual, audiovisual). We compared the mean difference between rhythmic and random sequences for the experimental (stroke patients) and control group (healthy subjects). The results show a significant mean difference between the sensory inputs presented in the rhythmic and random sequences in both groups (healthy and patients with stroke), and in all three experimental blocks (auditory, visual, and audiovisual), presenting higher values in rhythmic compared to random sequences in both healthy and patients with stroke group (see p values reported in Figure 9).

Significance Rhythmic - Random: control group							
		Random		Rhythmic		Comparison	
		M	SD	M	SD	Stat W	p value
Visual	q1	1.988	1.221	4.107	1.239	1	< .001
	q2	1.762	1.141	4.238	1.032	0	< .001
	q3	1.940	1.247	4.571	0.648	1	< .001
	q4	1.857	1.103	4.012	1.192	0	< .001
	q5	2.357	1.221	4.060	1.054	3	< .001
	q6	2.643	1.195	3.845	1.169	0	< .001
Audio	q1	2.012	1.231	3.524	1.483	7	0.001
	q2	2	1.220	3.536	1.419	13	0.001
	q3	2.024	1.072	3.845	1.182	10	< .001
	q4	2.048	1.195	3.298	1.387	9.5	0.001
	q5	2.298	1.158	3.321	1.154	12	0.004
	q6	2.643	1.244	3.381	1.094	22.5	0.010
Audiovisual	q1	2.131	1.198	3.679	1.238	2	< .001
	q2	1.857	1.276	3.893	1.244	1.5	< .001
	q3	1.821	1.215	4.298	0.640	0	< .001
	q4	1.940	1.101	3.5	1.257	5.5	< .001
	q5	2.298	1.221	3.643	1.166	6	0.001
	q6	2.726	1.235	3.667	1.218	14	0.005

Significance Rhythmic - Random: stroke group							
		Random		Rhythmic		Comparison	
		M	SD	M	SD	Stat W	p value
Visual	q1	2.726	1.416	4.024	0.997	16.5	0.003
	q2	2.476	1.483	4.036	1.088	16	0.002
	q3	2.655	1.354	4.488	0.718	0	< .001
	q4	2.429	1.381	3.810	1.112	9	0.002
	q5	3.107	1.279	4.298	0.875	2	0.001
	q6	3.321	1.282	4.369	0.843	4	0.001
Audio	q1	2.286	1.349	3.464	1.189	3.5	0.001
	q2	2.250	1.328	3.476	1.204	13.5	0.002
	q3	2.440	1.272	3.845	1.259	4	< .001
	q4	2.190	1.230	3.405	1.216	7	< .001
	q5	2.786	1.091	3.667	0.985	14	0.001
	q6	3.238	1.364	3.798	1.125	13	0.004
Audiovisual	q1	2.560	1.183	3.643	1.187	28.5	0.004
	q2	2.357	1.308	3.583	1.273	24.5	0.005
	q3	2.5	1.332	4.274	0.774	0	< .001
	q4	2.345	1.295	3.536	1.099	16.5	0.002
	q5	3.107	1.261	4.214	0.874	2	< .001
	q6	3.357	1.350	4.048	0.861	15	0.006

(a) Statical values for rhythmic- random comparison: control group (b) Statical values for rhythmic- random comparison: stroke group

Figure 9: Statistical values for different conditions, comparing the two population groups

Furthermore, Figures 10 represent the scores reported after each subjective question provided after the presentation of each sensory inputs in each experimental block, showing the differences between groups. From the statistical results reported in Table 11 we can observe that patients with stroke provided higher scores in the random sequences compared to the healthy subjects group. In detail, such difference was detected in Q1 and Q2 corresponding to the visual input and in Q3 related to the audiovisual input. However, no differences between groups were detected for the rhythmic sequences (see Table 11).

Moreover, we correlated the subjective report questions results obtained from Q2, after the observation of each sensory input presented in rhythmic sequences, with the mean power activity of the Occipital, Temporal and Sensorimotor ROIs, for each patient with stroke and healthy subjects group. The results from the correlation show that there was no significant relationship between the mean power spectrum activity and the Q2 score in any of the three different ROIs (temporal, sensorimotor, occipital). However, in the patients with stroke group we observed a positive correlation between the mean power activity in

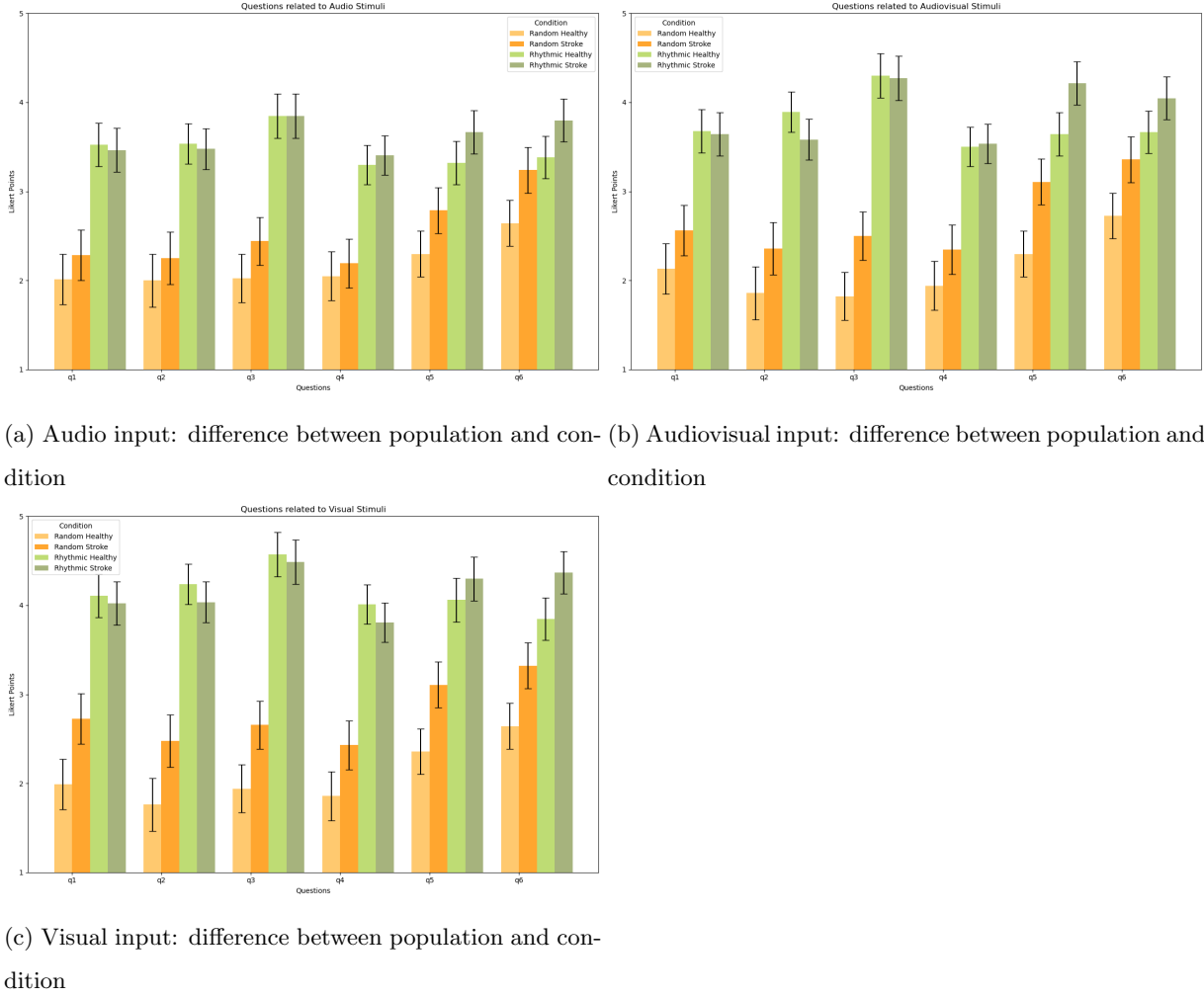


Figure 10: Bar plots representing difference between population and condition

Significance Control - Stroke group: Rhythmic							
		Control		Stroke		Comparison	
		<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	Stat <i>W</i>	<i>p</i> value
Visual	q1	4.02	1.00	4.11	1.24	87.50	0.76
	q2	4.04	1.09	4.24	1.03	66	0.39
	q3	4.49	0.72	4.57	0.65	42.50	0.52
	q4	3.81	1.11	4.01	1.19	81.50	0.59
	q5	4.30	0.88	4.06	1.05	69	0.29
	q6	4.37	0.84	3.85	1.17	62	0.11
Audio	q1	3.46	1.19	3.52	1.48	100	0.85
	q2	3.48	1.20	3.54	1.42	91.50	0.89
	q3	3.85	1.26	3.85	1.18	74.50	0.92
	q4	3.40	1.22	3.30	1.39	88.50	0.79
	q5	3.67	0.99	3.32	1.15	76	0.28
	q6	3.80	1.13	3.38	1.09	70	0.19
Audiovisual	q1	3.64	1.19	3.68	1.24	83	0.91
	q2	3.58	1.27	3.89	1.24	69.50	0.49
	q3	4.27	0.77	4.30	0.64	59.50	0.98
	q4	3.54	1.10	3.50	1.26	90.50	0.86
	q5	4.21	0.87	3.64	1.17	30	0.05
	q6	4.05	0.86	3.67	1.22	64.50	0.22

Significance Control - Stroke group: Random							
		Control		Stroke		Comparison	
		<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	Stat <i>W</i>	<i>p</i> value
Visual	q1	2.73	1.42	1.99	1.22	51.50	0.05
	q2	2.48	1.48	1.76	1.14	41.50	0.03
	q3	2.65	1.35	1.94	1.25	41.50	0.10
	q4	2.43	1.38	1.86	1.10	55.50	0.11
	q5	3.11	1.28	2.36	1.22	66.50	0.09
	q6	3.32	1.28	2.64	1.20	73.50	0.15
Audio	q1	2.29	1.35	2.01	1.23	92	0.63
	q2	2.25	1.33	2	1.22	74	0.62
	q4	2.19	1.23	2.05	1.20	82.50	0.90
	q5	2.79	1.09	2.30	1.16	63	0.20
	q6	3.24	1.36	2.64	1.24	60.50	0.16
	q1	2.56	1.18	2.13	1.20	62	0.18
Audiovisual	q2	2.36	1.31	1.86	1.28	37	0.06
	q3	2.50	1.33	1.82	1.21	33.50	0.04
	q4	2.35	1.30	1.94	1.10	56	0.20
	q5	3.11	1.26	2.30	1.22	47.50	0.06
	q6	3.36	1.35	2.73	1.23	84	0.29

(a) Statical values for rhythmic condition (b) Statical values for random condition

Figure 11: Statistical values for different conditions, comparing the two population groups

the Temporal ROI and the scoring collected in Q2, in the auditory experimental block (Tables 12 and 13).

	Correlations between subjective report question and neural activity: control group					
	OccipitalROI		TemporalROI		SensorimotorROI	
	<i>coefficient r</i>	<i>p value</i>	<i>coefficient r</i>	<i>p value</i>	<i>coefficient r</i>	<i>p value</i>
Audio	.021	.928	-.221	.335	-.104	.654
Audiovisual	-.388	.024	.919	.677	-.017	.941
Visual	-.295	.194	-.153	.507	-.317	.162

Figure 12: Correlation values: Spearman correlation coefficient r and p value

	Correlations between subjective report question and neural activity: stroke group					
	OccipitalROI		TemporalROI		SensorimotorROI	
	<i>coefficient r</i>	<i>p value</i>	<i>coefficient r</i>	<i>p value</i>	<i>coefficient r</i>	<i>p value</i>
Audio	.395	.076	.499	.021	.150	.516
Audiovisual	.052	.822	.103	.656	.070	.763
Visual	.375	.094	-.153	.507	-.126	.586

Figure 13: Correlation values: Spearman correlation coefficient r and p value

Finally, regarding the possible relationship between scoring collected in the Active-Q questionnaire (physical activity) and the mean average power in the sensorimotor ROI, the results did show any significant correlation in the healthy or patients with stroke group (Table 14).

	Correlations between brain and physical activity					
	Audio		Audiovisual		Visual	
Population	<i>coefficient r</i>	<i>p value</i>	<i>coefficient r</i>	<i>p value</i>	<i>coefficient r</i>	<i>p value</i>
Control	-.155	.504	-.294	.197	-.374	.095
Stroke	.071	.758	-.009	.969	.234	.308

Figure 14: Correlation values: Spearman's correlation coefficient r and p value

4. Discussion

The aim of the present study was to explore the brain dynamics supporting rhythmic sensorimotor synchronization when coupling auditory and visual inputs related to walking movement comparing healthy and stroke population, using an EEG frequency-tagging approach. In the present study we designed a task in which we presented an auditory (footstep sound) and visual (walking point-light figure) inputs at the frequency rate of 2 Hz in a rhythmic or random sequence, in three different blocks (auditory, visual and audiovisual). We chose to adopt the EEG technique with a frequency tagging approach, which has already given significative results to explore sensorimotor synchronization in both healthy (Cracco et al., 2022) and clinical populations, in particular with stroke patients (Nozaradan et al., 2017). Additionally, we decided to insert in the experiment some subjective report questions to inspect how the participants felt towards the stimuli presented, we aimed to know if they could feel relaxed or involved with the movement produced by sensory inputs. The results collected from these questions would have given us a clue on why brain activation would differ between distinct stimuli or population groups. All the EEG and subjective report results were compared between healthy subjects and patients to investigate the possible differences between them and the explanation standing behind them. As we commented before, in a previous study the authors have found that when presenting different sensory inputs (auditory, visual, audiovisual) related to the walking movement in rhythmic sequences at 2 Hz frequency rate, compared with 1 and 3.6 Hz, there was a higher synchronization with the sensorimotor cortex, with audio and audiovisual sensory inputs. In the current study, we confirmed the previous results in healthy subjects and in patients with stroke.

One of the main findings concerns higher synchronization effects in response to rhythmic auditory and the audiovisual stimuli, whereas not in response to the visual one. The results of the auditory input depict the synchronization effect at the temporal and sensorimotor cortex; while we can observe that the audiovisual inputs (which combined footsteps sounds and the walking point-light figure) showed the activation of the occipital cortex, together with the temporal and sensorimotor. Finally, in the visual input, only the activation of the occipital cortex could be found. These results were found in both healthy and stroke population (Figures 7 and 8). The crucial result of the induced higher synchronization effects within the sensorimotor cortex in the auditory input (footsteps sounds) compared to the visual input (walking point-light figure) can be explained through different previous findings. The auditory cortices can interact with subcortical areas that deal with the temporal processing, generating an internal periodic pulse-like beat when listening to auditory rhythms (Nozaradan et al., 2017). Sound entrains humans to move: neuroimaging studies have shown that rhythmic sequences activate motor cortical areas (Grahn & Brett, 2007). The difference between the synchronization in auditory and visual inputs can also be justified by the diverse role of attention in the auditory and visual system. In the last one it has been shown the key role of selective attention in visual search (Keshvari & Rosenholtz, 2016): there are elements that enhance this process, while others, such as biological motion and visual rhythms, that don't (Wolfe & Horowitz, 2017). Regarding this aspect, we can assume that it could be an explanation on why our visual sensory input (point-light figure) did not enhance selective attention and, consequently, not induced

sensorimotor synchronization as in the auditory inputs. These outcomes confirm earlier studies stating that the auditory stimuli enhance the temporal lobe and function as the best activator for sensorimotor entrainment (Thaut et al., 1999). On the opposite hand, we saw that the visual inputs did not seem to elicit neural entrainment, but that they only provoke the activation of the occipital lobe (Nehmad, 1998), responsible for sight.

Furthermore, we found that the EEG power amplitude recorded in stroke patients was weaker compared to the control group. The topographies of mean power activity in patients revealed the same activated ROIs across sensory inputs as those identified in the control group, but with an overall reduction. This weaker activation, particularly in the sensorimotor cortex, temporal, and occipital areas, is more evident when comparing the rhythmic and random conditions. While the stroke group still showed stronger activation in response to rhythmic stimuli, the difference between conditions highlights how diminished the activation of these ROIs truly is. This result confirms the outcomes found by another study with stroke patients and EEG frequency tagging approach (Nozaradan et al., 2017). Furthermore, research has showed that poor motor function is related with reduced sensory pathway information leading to a diminished amplitude activity power (Campfens et al., 2015).

Another important finding of this study is the higher sensorimotor synchronization in the rhythmic condition rather than in the random one. The EEG activity response to all the sensory inputs presented in a rhythmic or random sequence confirms the outcome of previous results, (e.g. Matamala-Gomez et al. in preparation; Haegens and Golumbic, 2018) showing a higher power amplitude in each region of interest for the stimuli in the rhythmic condition (see Topographies 7, 8). This can be explained by the need of brain oscillation to be synchronized to a repeated rhythm in order to enhance neural entrainment (Rosso et al., 2023). Moreover, as mentioned before, attention plays a key role for sensory integration: it has been observed that the perception of incoherent inputs can initially be represented by a unified stimulus (Bregman, 1990) and, consequently be ignored by the attentive system. This can disclose the higher synchronization with the sensory inputs repeated in the rhythmic sequence compared to the random one.

In addition to those findings, the analysis of the subjective report questions showed significantly higher values for the rhythmic sequences compared to the random sequences after the presentation of each sensory input in both healthy subjects and patients with stroke. From this result, we can hypothesize that during the presentation of the rhythmic sequences of the sensory input, the participants felt synchronized and more involved with the frequency rate of the sensory input (2Hz). This result is in line with previous studies (Nozaradan, Peretz, & Keller, 2016). From the reported results on the subjective questions, we also can state that in both groups, participants perceived the rhythmic stimuli as more enjoyable, and fluid compared to the random sequences.

Moreover, from the subjective questions we also have shown that participant perceived rhythmic sequences the participants experienced a sense of agency (Q4: *During the observation of the video, I felt like I was inside the body of the white dots (as if my own body was performing the movement)*, adapted for each sensory input) toward the observed point-light-figure walking movement in the visual and audiovisual experimental blocks and toward the footsteps sound in the auditory experimental block. This is an interesting result, as even we observed a coupling of the auditory input with the sensorimotor

cortex when presenting the auditory input in rhythmic sequences, we did not observe the same effect with the visual input presented in rhythmic sequences. However, different studies have shown that the sense of agency toward an action is related to activation with motor and sensorimotor areas (Buhrmann & Paolo, 2017; Longo & Haggard, 2009). Then, future studies can be directed in investigating whether providing a different visual input perspective or body morphology reproducing the walking movement at 2 Hz can synchronize with the sensorimotor cortex.

Even if previous studies have discussed that up to 70% of the stroke survivors can have impaired perception of different sensory stimuli, which may lead to misinterpretation of sensory information (C & Brady, 2022) and consequent lower brain activation, in the present study we did not show any issue in the integration or interpretation of the provided sensory inputs (walking point-light-figure and footstep sounds). However, from the topographies presented in figure 3.8, we also observe a lower synchronization effect with the sensorimotor cortex in patients with stroke compared to the healthy control group. In order to obtain a major knowledge about the possible differences in the synchronization of the different sensory inputs with the different brain ROIs between healthy and patients with stroke, further statistical analyses will be conducted. Further, we will investigate possible differences in sensory input integration in patients with cortical or subcortical injuries. It has been proven that the location of the lesion can influence EEG results (Clark, 2015), so a meticulous differentiation by this parameter, should be made.

5. Conclusion

The present study provides insights on brain dynamics underlying sensorimotor synchronization with different auditory, visual, and audiovisual inputs provided at 2Hz frequency by using an EEG frequency-tagging approach by means of neural entrainment. The neural oscillatory activity in both healthy subjects and patients with stroke showed higher sensorimotor synchronization in auditory and audiovisual stimuli rather than in the visual one. Further, sensorimotor synchronization, as well as temporal and occipital activation, were delineated with sensory stimuli repeated in a rhythmic sequence rather than a random one, emphasizing the sensorimotor synchronization when presenting the sensory input at frequency rate of 2 Hz. However, a lower synchronization in the sensorimotor, temporal, and occipital cortex has shown in the patients with stroke group. Such findings were also supported by the results of the subjective report questions which highlighted the positive feelings of fluidity and involvement with the movement in the rhythmic sequence, whereas these feelings were absent in the random condition. Further, participants also reported a higher sense of agency toward the auditory, visual, and audiovisual sensory inputs presented in rhythmic sequences.

Above all, the results of the present study constitute important progress in the knowledge on the role of the cortical cortex on the integration of auditory and visual walking inputs in patients with stroke. The results gave insights on the possibility of presenting the selected sensory inputs (auditory, visual, audiovisual), and in specific, auditory inputs, in rhythmic sequences at a specific frequency rate of 2Hz, when using sensory learning techniques for motor rehabilitation. In fact, previous studies have shown the effectiveness of using auditory input for walking rehabilitation, however the frequency rate presentation of the stimuli has not been specified before (Bolognini, Russo, & Edwards, 2016; Gomez-Andres et al., 2020). The results from this study can pave the way to new and effective motor rehabilitation techniques in patients presenting motor disorders.

A. Experimental Material

Table 1. Questions included in Active-Q^a and corresponding MET values.

Activity category	MET value
Daily occupation ^b	
Mostly sitting	1.5
A combination of sitting and standing up	2.3
Mostly standing up	3.0
Some physical activity	4.5
Heavy manual labor	6.0
Transportation	
Walking	4.0
Bicycling	4.0
By motorcycle or scooter	2.5
By car or taxi	1.0
By bus, train, subway, or boat	1.0
Leisure time activities	
Watching TV/DVDs	1.0
Using the computer	1.0
Sitting listening to music, sewing, etc	1.0
Playing a musical instrument or active computer games	2.0
Doing household chores	3.0
Shopping or other errands	2.3
Dancing	3.0
Walking	3.4
Bicycling	8.0
Regular sporting activities	
Aerobics	6.5
Weight lifting	6.0
Jogging or running	8.0
Athletics	6.0
Spinning	8.5
Swimming	6.0
Soccer, basketball, volleyball, or floorball (floor hockey)	6.0
Golf	4.5
Dance class	4.5
Horseback riding	4.0
Ice skating, ice hockey, or bandy	7.0
Skiing (downhill or cross country)	7.0
Martial arts	10.0
Boxing or wrestling	6.0
Tennis, badminton, or squash	7.0

Table 1: MET value labels (Bonn et al., 2012)

ID	Age	Gender	Affected extremity	Hemisphere injured
S_04	59	1	R	L
S_06	50	2	R	L
S_07	70	1	R	L
S_10	53	1	R	L
S_11	65	2	R	L
S_12	57	1	R	L
S_13	58	1	L	R
S_14	54	2	R	L
S_15	81	2	R	L
S_17	36	1	L	R
S_18	68	1	L	R
S_19	66	1	L	R
S_20	55	1	R	L
S_21	57	1	R	L
S_22	57	2	R	L
S_23	49	1	R	L
S_24	53	2	L	R
S_25	33	2	L	R
S_26	56	1	R	L
S_31	77	1	L and R	R and L
S_33	30	1	L	R

Table 2: Database for stroke population (Gender: 1 = man; 2 = woman; Hemispheres: R = right; L = left)

A.1 Active-Q questionnaire

Active-Q

Revised version
December 2011

Active-Q – A Physical Activity Questionnaire for Adults

The following questions concern your physical activity during the last year

Physical activity level – daily occupation

What is your normal activity level at your daily occupation (work, studies or equivalent)?

Mostly Sitting	A combination of sitting and standing up	Mostly standing up	Some physical activity	Heavy manual labor
☐	☐	☐	☐	☐

Duration – daily occupation

Approximately how many hours per week do you normally conduct your daily occupation (work, studies or equivalent)?

- ☐ 5 hours or less
☐ 6 - 10 hours
☐ 11 – 20 hours
☐ 21 – 30 hours
☐ 31 – 40 hours
☐ 41 – 50 hours
☐ More than 50 hours
☐ Don't know / Don't want to answer

Means of transportation to daily occupation

How do you normally get to and from work, studies or other daily occupation?

- ☐ Walking ☐ By bus, train, subway or boat
☐ Cycling ☐ Other
☐ By motorcycle, moped or scooter ☐ Does not apply to me
☐ By car or taxi ☐ Don't know / Don't want to answer

Follow-up questions regarding means of transportation.

Selected activities in the previous screening question are presented to the respondent

How often do you get to your daily occupation in the following ways and approximately how long does it take?

	Number of days per week	Time, one way per day
Walking	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7	☐ Less than 15 minutes ☐ 15 - 29 minutes ☐ 30 - 44 minutes ☐ 45 - 59 minutes ☐ 1 - 2 hours ☐ Longer than 2 hours

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	Number of days per week	Time, one way per day
Cycling	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7	☐ Less than 15 minutes ☐ 15 - 29 minutes ☐ 30 - 44 minutes ☐ 45 - 59 minutes ☐ 1 - 2 hours ☐ Longer than 2 hours
By motorcycle, moped or scooter	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7	☐ Less than 15 minutes ☐ 15 - 29 minutes ☐ 30 - 44 minutes ☐ 45 - 59 minutes ☐ 1 - 2 hours ☐ Longer than 2 hours
By car or taxi	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7	☐ Less than 15 minutes ☐ 15 - 29 minutes ☐ 30 - 44 minutes ☐ 45 - 59 minutes ☐ 1 - 2 hours ☐ Longer than 2 hours
By bus, train, subway or boat	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7	☐ Less than 15 minutes ☐ 15 - 29 minutes ☐ 30 - 44 minutes ☐ 45 - 59 minutes ☐ 1 - 2 hours ☐ Longer than 2 hours
Other	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7	☐ Less than 15 minutes ☐ 15 - 29 minutes ☐ 30 - 44 minutes ☐ 45 - 59 minutes ☐ 1 - 2 hours ☐ Longer than 2 hours

Leisure time activities

Which of the following activities do you engage in at least once per week in your leisure time?

- | | |
|---|--|
| ☐ Watching TV/DVDs etc. | ☐ Dancing |
| ☐ Using the computer, reading emails, playing computer games, play station, X-box etc. | ☐ Walking, Nordic walking, walking a dog (not as transport to daily occupation) |
| ☐ Sitting reading, writing, sewing etc. | ☐ Bicycling (not as transport to daily occupation) |
| ☐ Playing a musical instrument or active computer games (e.g. Wii) | ☐ None of the above |
| ☐ Doing household chores (e.g. cleaning, laundry, child care, gardening etc.) | ☐ Don't know /Don't want to answer |
| ☐ Shopping, or other errands | |

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How many days per week do you dedicate time to the following leisure time activities and how much time do you spend per day on the activity?

	Number of days per week	Total time per day
Watching TV/DVDs	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 1 - 2 hours ☐ 3 - 4 hours ☐ 5 - 8 hours ☐ Longer than 8 hours
Using the computer	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 1 - 2 hours ☐ 3 - 4 hours ☐ 5 - 8 hours ☐ Longer than 8 hours
Sitting writing, reading, sewing etc.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 1 - 2 hours ☐ 3 - 4 hours ☐ 5 - 8 hours ☐ Longer than 8 hours]
Playing a musical instrument or active computer games (e.g. Wii)	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 1 - 2 hours ☐ 3 - 4 hours ☐ 5 - 8 hours ☐ Longer than 8 hours
Doing household chores, cleaning, doing laundry, taking care of children, garden work etc.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 1 - 2 hours ☐ 3 - 4 hours ☐ 5 - 8 hours ☐ Longer than 8 hours
Shopping or other errands	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 1 - 2 hours ☐ 3 - 4 hours ☐ 5 - 8 hours ☐ Longer than 8 hours

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	Number of days per week	Total time per day
Dancing (e.g. disco or ball room)	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 1 - 2 hours ☐ 3 - 4 hours ☐ 5 - 8 hours ☐ Longer than 8 hours
Walking (not as transport to daily occupation), Nordic walking or walking a dog	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7	☐ Less than 15 minutes ☐ 15 – 29 minutes ☐ 30 – 44 minutes ☐ 45 – 59 minutes ☐ 1 – 2 hours ☐ Longer than 2 hours
Cycling (not as transport to daily occupation)	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7	☐ Less than 15 minutes ☐ 15 – 29 minutes ☐ 30 – 44 minutes ☐ 45 – 59 minutes ☐ 1 – 2 hours ☐ Longer than 2 hours

Overall sporting activity or exercise

Do you exercise or participate in any sporting activities regularly?

- ☐ Yes
☐ No
☐ Don't know / Don't want to answer

Sporting activities

What kind of exercise or sports activities do you participate in? If your type of activity is not included in the list below, please choose the most similar one.

- ☐ Aerobics or cardio fitness class
☐ Weight lifting
☐ Jogging, running or orienteering
☐ Athletics (e.g. high jump, long jump or three-step)
☐ Spinning or cycling in demanding terrain
☐ Swimming
☐ Ball sports (e.g. soccer, basketball, volley ball or floorball)
☐ Golf

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- ☐ Dance class or competitive dancing (e.g. ballroom, ballet, jazz or street)
☐ Horseback riding
☐ Skating, ice hockey or bandy
☐ Skiing downhill or cross country
☐ Martial arts (e.g. judo or karate)
☐ Boxing or wrestling
☐ Tennis, badminton or table tennis
☐ Squash
☐ Sailing, surfing, canoeing or rowing
☐ Motor sports (e.g. motocross)
☐ Rock climbing
☐ Yoga, Pilates or Tai chi
☐ Other activity or sport (please state) _____
☐ None of these
☐ Don't know / Don't want to answer

Follow-up questions regarding regular sporting activities**Selected activities in the previous screening question are presented to the respondent**

How often do you participate in the following sports or the following activities and for how long each time?

	Number of times	Time per session
Aerobics or cardio fitness class	☐ 1 - 3 times a month ☐ Once a week ☐ 2 - 3 times a week ☐ 4 - 5 times a week ☐ 6 - 7 times a week	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 60 - 119 minutes ☐ 2 - 4 hours ☐ More than 4 hours
Weight lifting	☐ 1 - 3 times a month ☐ Once a week ☐ 2 - 3 times a week ☐ 4 - 5 times a week ☐ 6 - 7 times a week	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 60 - 119 minutes ☐ 2 - 4 hours ☐ More than 4 hours
Jogging, running or orienteering	☐ 1 - 3 times a month ☐ Once a week ☐ 2 - 3 times a week ☐ 4 - 5 times a week ☐ 6 - 7 times a week	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 60 - 119 minutes ☐ 2 - 4 hours ☐ More than 4 hours
Athletics (e.g. high jump, long jump or three-step)	☐ 1 - 3 times a month ☐ Once a week ☐ 2 - 3 times a week ☐ 4 - 5 times a week ☐ 6 - 7 times a week	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 60 - 119 minutes ☐ 2 - 4 hours ☐ More than 4 hours
Spinning or cycling in demanding terrain	☐ 1 - 3 times a month ☐ Once a week ☐ 2 - 3 times a week ☐ 4 - 5 times a week ☐ 6 - 7 times a week	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 60 - 119 minutes ☐ 2 - 4 hours ☐ More than 4 hours
Swimming	☐ 1 - 3 times a month ☐ Once a week	☐ Less than 30 minutes ☐ 30 - 59 minutes

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	Number of times	Time per session
	☐ 2 - 3 times a week ☐ 4 - 5 times a week ☐ 6 - 7 times a week	☐ 60 - 119 minutes ☐ 2 - 4 hours ☐ More than 4 hours
Ball sports (e.g. soccer, basketball, volley ball or floor ball)	☐ 1 - 3 times a month ☐ Once a week ☐ 2 - 3 times a week ☐ 4 - 5 times a week ☐ 6 - 7 times a week	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 60 - 119 minutes ☐ 2 - 4 hours ☐ More than 4 hours
Golf	☐ 1 - 3 times a month ☐ Once a week ☐ 2 - 3 times a week ☐ 4 - 5 times a week ☐ 6 - 7 times a week	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 60 - 119 minutes ☐ 2 - 4 hours ☐ More than 4 hours
Dance class or competitive dancing	☐ 1 - 3 times a month ☐ Once a week ☐ 2 - 3 times a week ☐ 4 - 5 times a week ☐ 6 - 7 times a week	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 60 - 119 minutes ☐ 2 - 4 hours ☐ More than 4 hours
Horseback riding	☐ 1 - 3 times a month ☐ Once a week ☐ 2 - 3 times a week ☐ 4 - 5 times a week ☐ 6 - 7 times a week	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 60 - 119 minutes ☐ 2 - 4 hours ☐ More than 4 hours
Skating, ice-hockey or bandy	☐ 1 - 3 times a month ☐ Once a week ☐ 2 - 3 times a week ☐ 4 - 5 times a week ☐ 6 - 7 times a week	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 60 - 119 minutes ☐ 2 - 4 hours ☐ More than 4 hours
Skiing, downhill or cross country	☐ 1 - 3 times a month ☐ Once a week ☐ 2 - 3 times a week ☐ 4 - 5 times a week ☐ 6 - 7 times a week	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 60 - 119 minutes ☐ 2 - 4 hours ☐ More than 4 hours
Martial arts (e.g. judo or karate)	☐ 1 - 3 times a month ☐ Once a week ☐ 2 - 3 times a week ☐ 4 - 5 times a week ☐ 6 - 7 times a week	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 60 - 119 minutes ☐ 2 - 4 hours ☐ More than 4 hours
Boxing or wrestling	☐ 1 - 3 times a month ☐ Once a week ☐ 2 - 3 times a week ☐ 4 - 5 times a week ☐ 6 - 7 times a week	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 60 - 119 minutes ☐ 2 - 4 hours ☐ More than 4 hours
Tennis, badminton or table tennis	☐ 1 - 3 times a month ☐ Once a week ☐ 2 - 3 times a week ☐ 4 - 5 times a week ☐ 6 - 7 times a week	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 60 - 119 minutes ☐ 2 - 4 hours ☐ More than 4 hours
Squash	☐ 1 - 3 times a month	☐ Less than 30 minutes

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	Number of times	Time per session
	☐ Once a week ☐ 2 - 3 times a week ☐ 4 - 5 times a week ☐ 6 - 7 times a week	☐ 30 - 59 minutes ☐ 60 - 119 minutes ☐ 2 - 4 hours ☐ More than 4 hours
Sailing, surfing, canoeing or rowing	☐ 1 - 3 times a month ☐ Once a week ☐ 2 - 3 times a week ☐ 4 - 5 times a week ☐ 6 - 7 times a week	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 60 - 119 minutes ☐ 2 - 4 hours ☐ More than 4 hours
Motor sports (e.g. motocross)	☐ 1 - 3 times a month ☐ Once a week ☐ 2 - 3 times a week ☐ 4 - 5 times a week ☐ 6 - 7 times a week	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 60 - 119 minutes ☐ 2 - 4 hours ☐ More than 4 hours
Rock climbing	☐ 1 - 3 times a month ☐ Once a week ☐ 2 - 3 times a week ☐ 4 - 5 times a week ☐ 6 - 7 times a week	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 60 - 119 minutes ☐ 2 - 4 hours ☐ More than 4 hours
Yoga, Pilates or Tai chi	☐ 1 - 3 times a month ☐ Once a week ☐ 2 - 3 times a week ☐ 4 - 5 times a week ☐ 6 - 7 times a week	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 60 - 119 minutes ☐ 2 - 4 hours ☐ More than 4 hours
Other (displayed automatically)	☐ 1 - 3 times a month ☐ Once a week ☐ 2 - 3 times a week ☐ 4 - 5 times a week ☐ 6 - 7 times a week	☐ Less than 30 minutes ☐ 30 - 59 minutes ☐ 60 - 119 minutes ☐ 2 - 4 hours ☐ More than 4 hours

Do you compete regularly in a sport?**Follow-up question if respondent reported regular physical activity**

- ☐ Yes
☐ No
☐ Don't know / Don't want to answer

How many hours do you usually sleep during a typical night?

- ☐ Less than 5 hours ☐ 9 hours
☐ 6 hours ☐ 10 hours or longer
☐ 7 hours ☐ Don't know / Don't want to answer
☐ 8 hours

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